

Business Case Study

Yulu - Hypothesis Testing

Submitted By: Heena Khan

Colab Link:

https://colab.research.google.com/drive/1uWdnMQLrvO_I3nJG5GiVNeu4Q53kWpQI?usp=sharing

About Yulu

Yulu is India's leading micro-mobility service provider, which offers unique vehicles for the daily commute. Starting off as a mission to eliminate traffic congestion in India, Yulu provides the safest commute solution through a user-friendly mobile app to enable shared, solo and sustainable commuting.

Yulu zones are located at all the appropriate locations (including metro stations, bus stands, office spaces, residential areas, corporate offices, etc) to make those first and last miles smooth, affordable, and convenient!

Yulu has recently suffered considerable dips in its revenues. They have contracted a consulting company to understand the factors on which the demand for these shared electric cycles depends. Specifically, they want to understand the factors affecting the demand for these shared electric cycles in the Indian market.

How you can help here?

The company wants to know:

- Which variables are significant in predicting the demand for shared electric cycles in the Indian market?
- How well those variables describe the electric cycle demands

Dataset:

Dataset Link: yulu_data.csv

Column Profiling:

- datetime: datetime
- season: season (1: spring, 2: summer, 3: fall, 4: winter)
- holiday: whether day is a holiday or not (extracted from <http://dchr.dc.gov/page/holiday-schedule>)
- workingday: if day is neither weekend nor holiday is 1, otherwise is 0.
- weather:
 - 1: Clear, Few clouds, partly cloudy, partly cloudy
 - 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
 - 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
 - 4: Heavy Rain + Ice Pellets + Thunderstorm + Mist, Snow + Fog
- temp: temperature in Celsius
- atemp: feeling temperature in Celsius
- humidity: humidity
- windspeed: wind speed
- casual: count of casual users
- registered: count of registered users
- count: count of total rental bikes including both casual and registered

Concept Used:

- Bi-Variate Analysis
- 2-sample t-test: testing for difference across populations
- ANNOVA
- Chi-square

Problem Statement:

Yulu wants to identify:

1. Which variables significantly affect the demand for shared electric cycles.
2. How well these variables explain the rental demand.

The target variable is:

- **count** : Total number of rented cycles

Independent variables include:

- season
- holiday
- workingday
- weather
- temp
- atemp
- humidity
- windspeed

Import Required Libraries :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import ttest_ind # T-test for independent samples
from scipy.stats import shapiro # Shapiro-Wilk's test for Normality
from scipy.stats import levene # Levene's test for Equality of Variance
from scipy.stats import f_oneway # One-way ANOVA
from scipy.stats import kruskal
from scipy.stats import chi2_contingency # Chi-square test of
independence
```

Load Dataset :

```
df=pd.read_csv("/content/sample_data/bike_sharing.csv")
```

Basic Exploratory Data Analysis (EDA) :

View Dataset

```
df.head()
```


	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count
mean	2.506614	0.028569	0.680875	1.418427	20.23086	23.655084	61.886460	12.799395	36.021955	155.552177	191.574132
std	1.116174	0.166599	0.466159	0.633839	7.79159	8.474601	19.245033	8.164537	49.960477	151.039033	181.144454
min	1.000000	0.000000	0.000000	1.000000	0.82000	0.76000	0.000000	0.000000	0.000000	0.000000	1.000000
25%	2.000000	0.000000	0.000000	1.000000	13.94000	16.665000	47.000000	7.001500	4.000000	36.000000	42.000000
50%	3.000000	0.000000	1.000000	1.000000	20.50000	24.240000	62.000000	12.998000	17.000000	118.000000	145.000000
75%	4.000000	0.000000	1.000000	2.000000	26.24000	31.060000	77.000000	16.997900	49.000000	222.000000	284.000000
max	4.000000	1.000000	1.000000	4.000000	41.00000	45.455000	100.000000	56.996900	367.000000	886.000000	977.000000

```
df.describe(include="O")
```

	datetime
count	10886
unique	10886
top	19-12-2012 23.00
freq	1

Missing Values

```
df.isnull().sum()
```

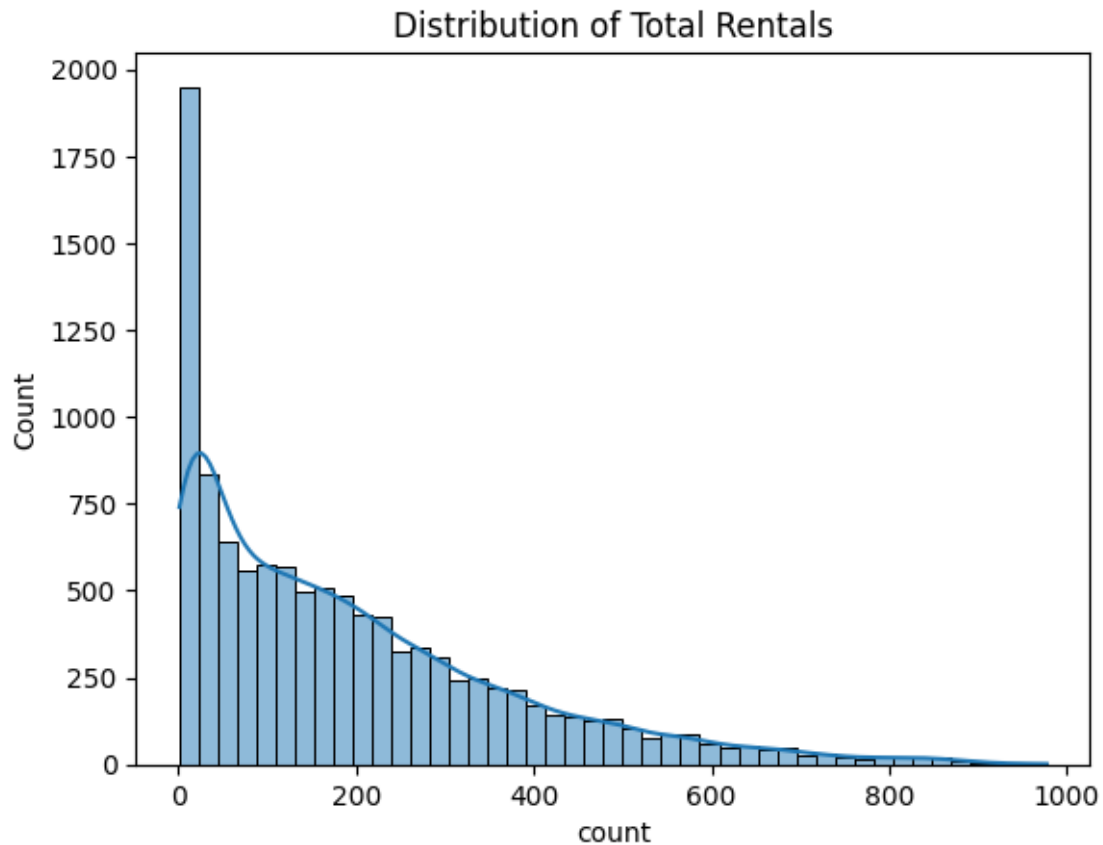
	0
datetime	0
season	0
holiday	0
workingday	0
weather	0
temp	0
atemp	0
humidity	0
windspeed	0
casual	0
registered	0
count	0

dtype: int64

Univariate Analysis :

Distribution of Count

```
sns.histplot(df['count'], kde=True)
plt.title("Distribution of Total Rentals")
plt.show()
```



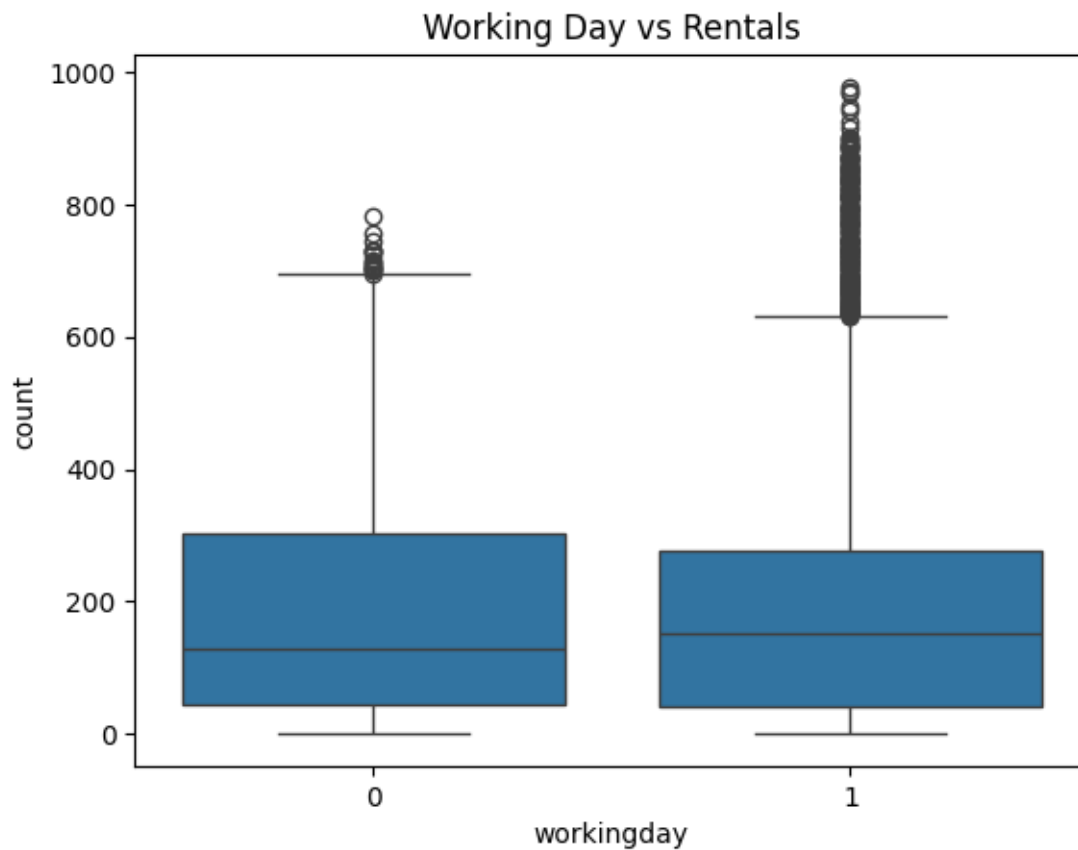
Observation

- Rental count is right-skewed
- Most rentals are concentrated in lower-to-medium range

Bivariate Analysis

Working Day vs Count

```
sns.boxplot(x='workingday', y='count', data=df)
plt.title("Working Day vs Rentals")
plt.show()
```

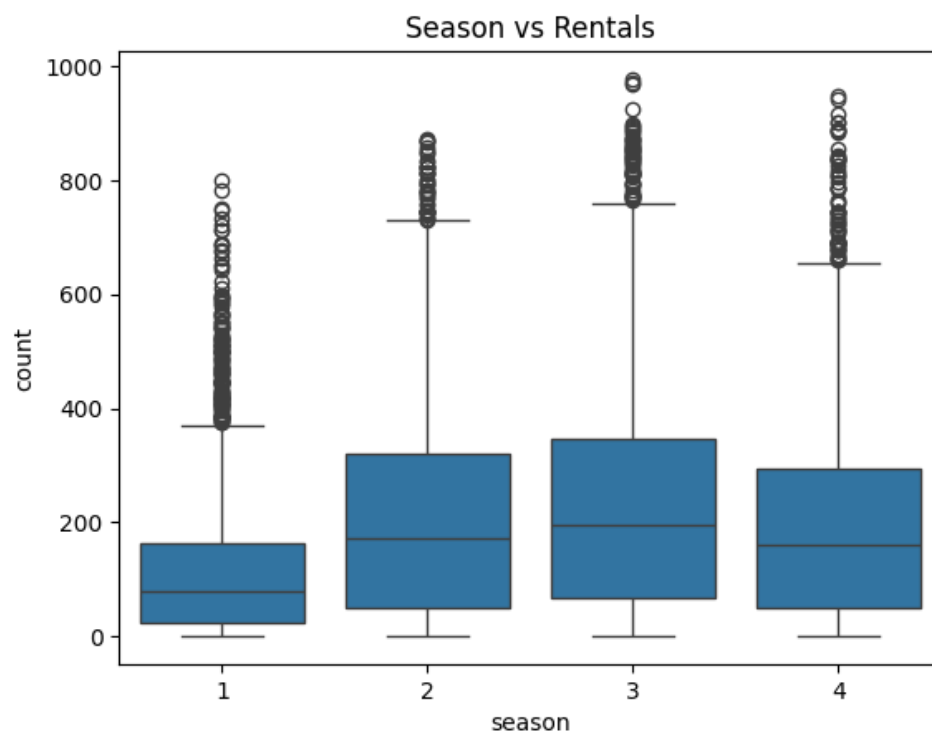


Observation

- Rentals appear slightly higher on working days

Season vs Count

```
sns.boxplot(x='season', y='count', data=df)
plt.title("Season vs Rentals")
plt.show()
```

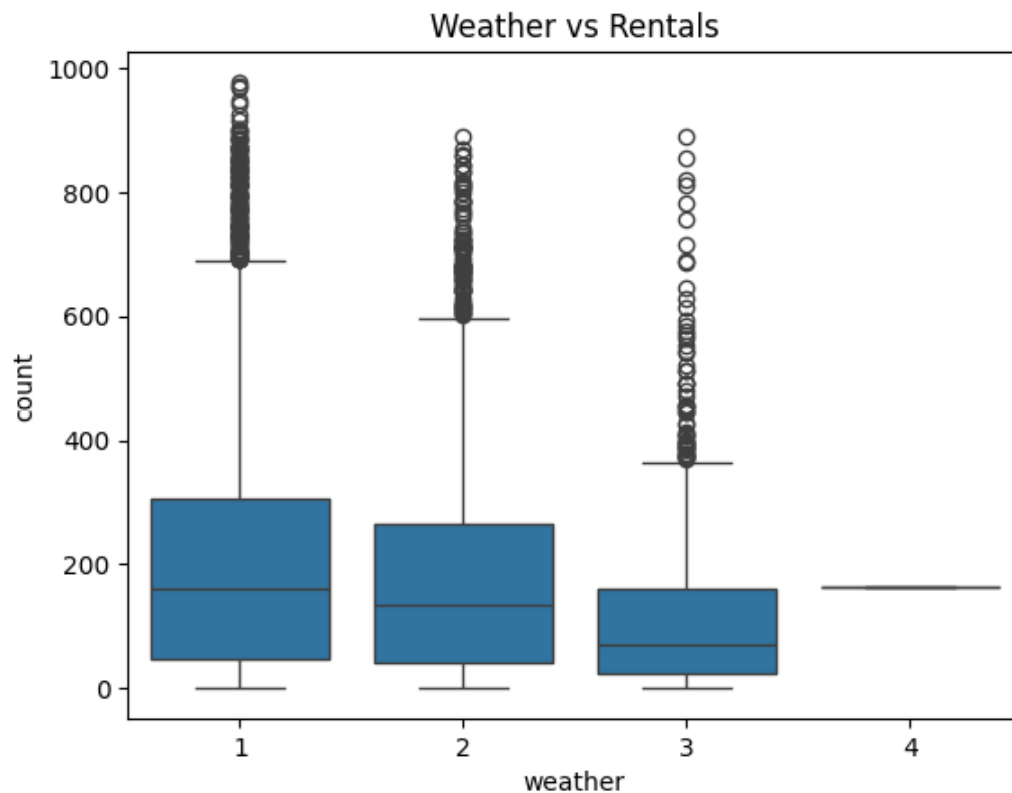


Observation

- Fall season shows highest rentals
- Spring season shows lowest rentals

Weather vs Count

```
sns.boxplot(x='weather', y='count', data=df)
plt.title("Weather vs Rentals")
plt.show()
```



Observation

- Clear weather has highest rentals
- Bad weather significantly reduces rentals

```
def dist_check(df,col_name):
    print("Unique values",df[col_name].unique())
    print("Value coounts", df[col_name].value_counts())
```

```
col_list=["season","holiday","workingday","weather"]
for items in col_list:
    print(f"Column Name: {items}")
    dist_check(df,col_name=items)
    print("-----"*10)
```

```
Column Name: season
Unique values [1 2 3 4]
Value coounts season
4      2734
2      2733
3      2733
1      2686
```

Name: count, dtype: int64

Column Name: holiday

Unique values [0 1]

Value counts holiday

0 10575

1 311

Name: count, dtype: int64

Column Name: workingday

Unique values [0 1]

Value counts workingday

1 7412

0 3474

Name: count, dtype: int64

Column Name: weather

Unique values [1 2 3 4]

Value counts weather

1 7192

2 2834

3 859

4 1

Name: count, dtype: int64

```
df_only_num=df.select_dtypes(include=["float","int"])
```

```
plt.figure(figsize=(12, 7)) ## This define my canvas size.
sns.heatmap(df_only_num.corr(method='spearman'),
            annot=True, cmap='viridis')
```

<Axes: >

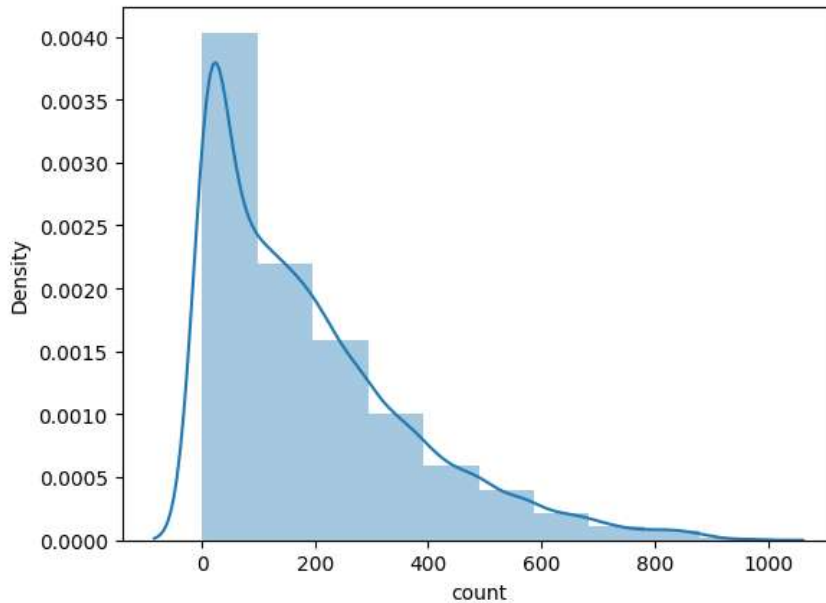


```
dfn=df.drop(columns=['casual','registered','atemp'])
dfn.head()
```

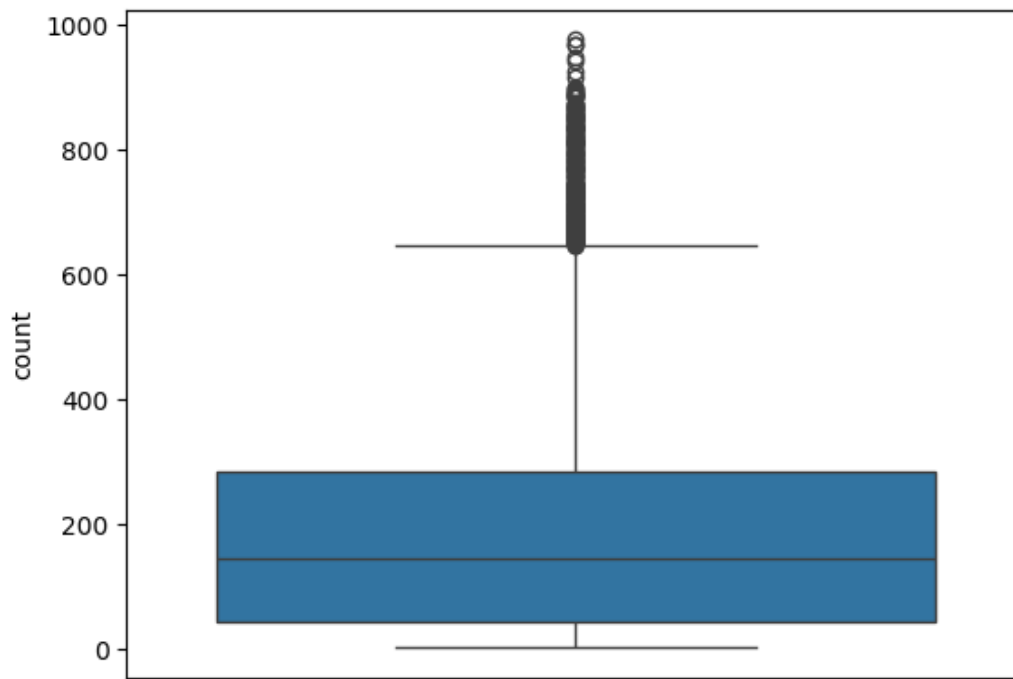
	datetime	season	holiday	workingday	weather	temp	humidity	windspeed	count
0	01-01-2011 0.00	1	0	0	1	9.84	81	0.0	16

	datetime	season	holiday	workingday	weather	temp	humidity	windspeed	count
1	01-01-2011 1.00	1	0	0	1	9.02	80	0.0	40
2	01-01-2011 2.00	1	0	0	1	9.02	80	0.0	32
3	01-01-2011 3.00	1	0	0	1	9.84	75	0.0	13
4	01-01-2011 4.00	1	0	0	1	9.84	75	0.0	1

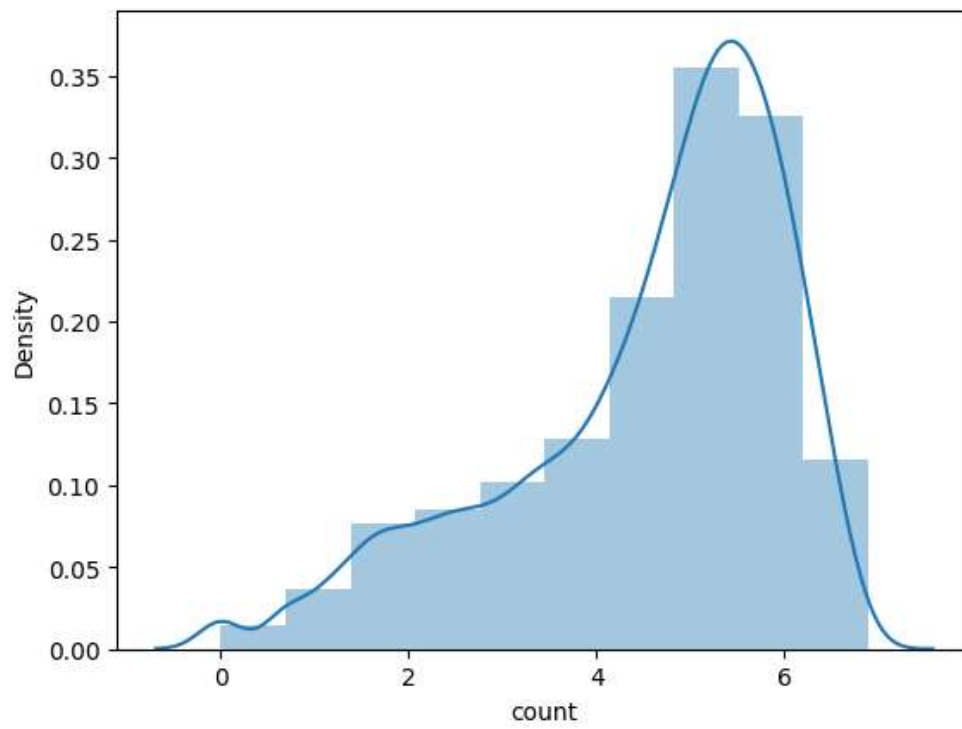
```
sns.distplot(dfn['count'],bins=10)
```



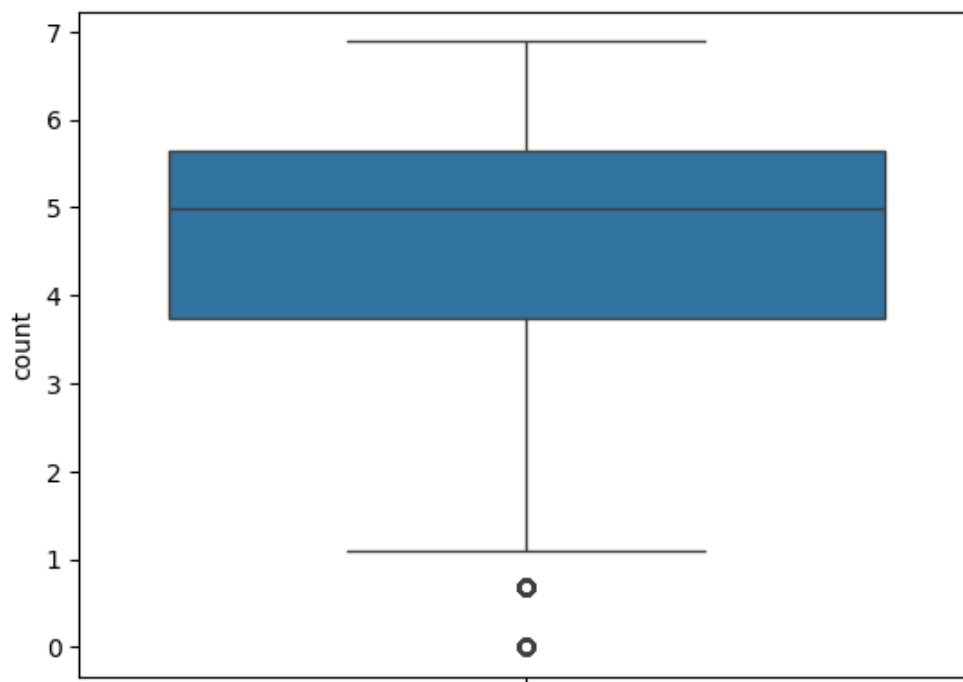
```
sns.boxplot(y=dfn['count'])
plt.show()
```



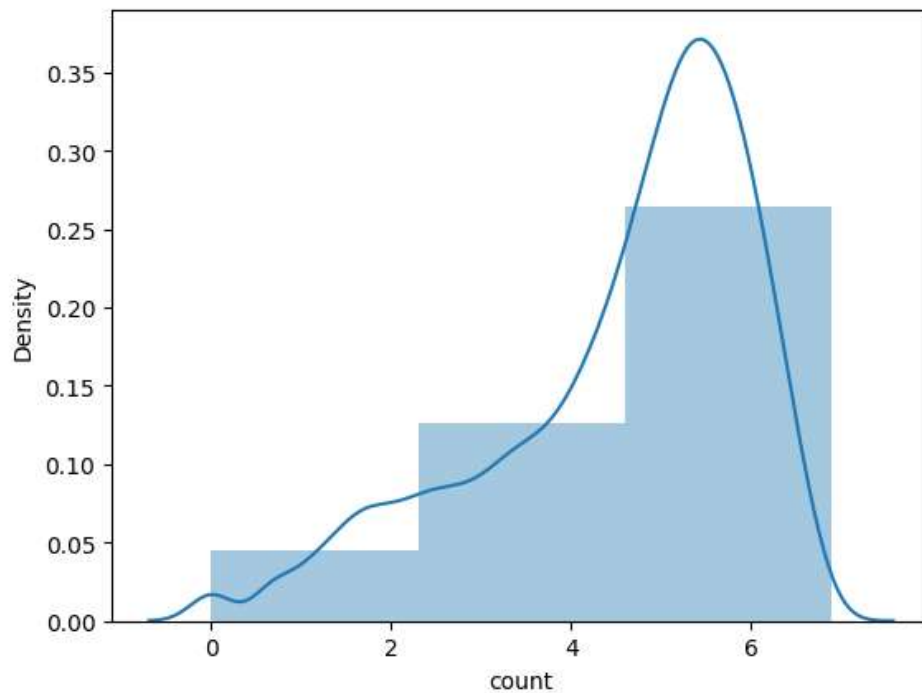
```
sns.distplot(np.log(dfn['count']),bins=10)
```



```
sns.boxplot(y=np.log(dfn['count']))
plt.show()
```



```
sns.distplot(np.log(dfn['count']),bins=3)
```



Statistical Hypothesis Testing

Is there any significant difference between the no. of bike rides on weekdays and weekends?

H_0 : The demand of bikes on weekdays is greater or similar to the demand of bikes on weekend.

H_a : The demand of bikes on weekdays is less than the demand of bikes on weekend.

Let μ_1 and μ_2 be the average no. of bikes rented on weekdays and weekends respectively.

Mathematically, the above formulated hypothesis can be written as:

$H_0: \mu_1 \geq \mu_2$

$H_a: \mu_1 < \mu_2$

```
dfn.groupby('workingday')['count'].size()
```

	count
workingday	
0	3474
1	7412

dtype: int64

```
weekday=dfn[dfn['workingday']==1]['count'].sample(3000)
```

```
weekend=dfn[dfn['workingday']==0]['count'].sample(3000)
```

```
round(weekday.std(), 2)
```

182.41

```
round(weekend.std(), 2)
```

```
174.4
```

```
alpha=0.05
```

```
def result(p_value, alpha):  
    if p_value < alpha:  
        print(f'As the p-value {p_value} is less than the level of  
significance, we reject the null hypothesis.')  
    else:  
        print(f'As the p-value {p_value} is greater than the level of  
significance, we fail to reject the null hypothesis.')
```

```
test_stat,p_value=ttest_ind(weekday,weekend,equal_var=False,alternative  
="less")
```

```
p_value
```

```
np.float64(0.5708799915055375)
```

```
result(p_value,alpha)
```

As the p-value 0.5708799915055375 is greater than the level of significance, we fail to reject the null hypothesis.

Is the demand of bicycles on rent same for different weather conditions?

```
dfn.groupby('weather')['count'].describe()
```

	count	mean	std	min	25%	50%	75%	max
weather								
1	7192.0	205.236791	187.959566	1.0	48.0	161.0	305.0	977.0
2	2834.0	178.955540	168.366413	1.0	41.0	134.0	264.0	890.0
3	859.0	118.846333	138.581297	1.0	23.0	71.0	161.0	891.0
4	1.0	164.000000	NaN	164.0	164.0	164.0	164.0	164.0

Step 1: Define the null and alternate hypothesis

H₀: The average no. of bike rides in different weather conditions are equal.

H_a: The average no. of bike rides in different weather conditions are not equal.

Let μ_1 and μ_2 be the average no. of bikes rented on weekdays and weekends respectively.

```
dfn=dfn[~(dfn['weather']==4)]
```

```
w1=dfn[dfn['weather']==1]['count'].sample(750)
w2=dfn[dfn['weather']==2]['count'].sample(750)
w3=dfn[dfn['weather']==3]['count'].sample(750)
#w4=dfn[dfn['weather']==4]['count'].sample(750)
#
```

Shapiro test

```
## Ho : Normal Ha: Npt normal

w,p_value=shapiro(dfn['count'].sample(5000))
result(p_value,alpha)
```

As the p-value 6.9702886960469e-53 is less than the level of significance, we reject the null hypothesis.

```
stat,p_value=kruskal(w1,w2,w3)
result(p_value,alpha)
```

As the p-value 2.851460964932703e-26 is less than the level of significance, we reject the null hypothesis.

Are the weather conditions significantly different during different seasons?

Step 1: Define the null and alternate hypothesis

H0: Weather conditions are independent of the season.

Ha: Weather condition depends on the ongoing season.

```
dict1={1:'Sunny',
        2:'Cloudy',
        3:'Rainy'}

dfn["weather_enc"]=dfn["weather"].map(dict1)
```

```
dict2={1:'Summer',
        2:'Monsoon',
        3:'Winter',
        4:'Autumn'}

dfn["season_enc"]=dfn["season"].map(dict2)
```

```
contingency=pd.crosstab(dfn.season_enc,dfn.weather_enc)
```

weather_enc	Cloudy	Rainy	Sunny
season_enc			
Autumn	807	225	1702
Monsoon	708	224	1801
Summer	715	211	1759
Winter	604	199	1930

```
chi2, pval, dof, exp_freq = chi2_contingency(contingency,
correction=False)
print('Chi-square Statistic: {} \n P-value: {} \n Degree of Freedom: {}
\n Expected Frequencies: {}'.format(chi2, pval, dof, exp_freq))
```

```
Chi-square Statistic: 46.10145731073249
P-value: 2.8260014509929343e-08
Degree of Freedom: 6
Expected Frequencies: [[ 711.81956821  215.75617823 1806.42425356]
[ 711.55920992  215.67726229 1805.76352779]
[ 699.06201194  211.8892972  1774.04869086]
[ 711.55920992  215.67726229 1805.76352779]]
```

```
result(pval,alpha)
```

As the p-value 2.8260014509929343e-08 is less than the level of significance, we reject the null hypothesis.

EDA based insights -

Total 10,886 rows were present in the data set.

Neither missing values, nor duplicate rows were found.

'temp' and 'atemp' columns were found to be highly correlated.

Dropping one of them (atemp) to avoid multicollinearity.

'count', 'casual' and 'registered' columns were highly correlated.

Dropping casual & registered columns to avoid multicollinearity.

Outlier values were found in the 'count' column.

Insights from hypothesis testing -

The no. of bikes rented on weekdays is comparatively higher than on weekends.

The no. of bikes rented on regular days is comparatively higher than on holidays.

The demand of bicycles on rent differs under different weather conditions.

The demand of bicycles on rent is different during different seasons.

The weather conditions are surely dependent upon the ongoing season.

Miscellaneous observations -

The distribution of 'count' column wasn't actually normal or near normal.

Infact the column's distribution is found to be a bit skewed towards right.

Generic recommendations -

The demand of bikes on rent are usually higher during Weekdays.

The demand of bikes on rent are usually higher during Regular days.

The chances of person renting a bike are usually higher during Season 3.

The chances of person renting a bike are usually higher during Weather condition 1.

Business Recommendations for Yulu

Increase Supply During High Demand

- Deploy more cycles during:
 - Pleasant weather
 - Office hours
 - Summer/fall seasons

Weather-Based Dynamic Planning

- Reduce deployment during heavy rain or storms
- Introduce weather-based pricing/promotions

Corporate Commuter Focus

- Since working days show higher rentals:
 - Partner with IT parks
 - Expand metro connectivity zones

We recommend the company to maintain the bike stocks accordingly.